CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

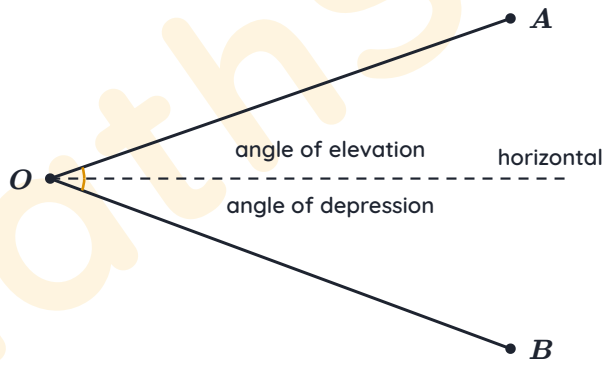
Chapter 22: Heights and Distances

Topic-wise short notes with solved examples

Trigonometric ratios are used to find heights and distances that cannot be measured directly. Every problem comes down to one or two right-angled triangles.

1 Angles of elevation and depression

- The **line of sight** is the line from the observer's eye to the object.
- The **angle of elevation** is the angle between the horizontal and the line of sight when the object is **above** the level of the eye.
- The **angle of depression** is the angle between the horizontal and the line of sight when the object is **below** the level of the eye.
- The angle of depression of B from A equals the angle of elevation of A from B (alternate angles).



Both angles are measured from the horizontal through the eye O

REMEMBER

Both angles are measured from the horizontal line through the observer's eye, never from the vertical.

2 Which ratio to use

Known and required sides	Ratio
Height and horizontal distance	$\tan$
Height and slant length (ladder, rope, string)	$\sin$
Horizontal distance and slant length	$\cos$

- Standard values: $\tan 30^\circ = \dfrac{1}{\sqrt{3}}$, $\tan 45^\circ = 1$, $\tan 60^\circ = \sqrt{3}$, $\sin 30^\circ = \cos 60^\circ = \dfrac{1}{2}$, $\sin 60^\circ = \cos 30^\circ = \dfrac{\sqrt{3}}{2}$.
- For other angles use the tables. Give the answer to the accuracy asked: nearest metre, or a number of significant figures.

3 Problems with one triangle

Example 3.1 The angle of elevation of the top of a tower from a point 40 m from its foot is 60°. Find the height of the tower, correct to the nearest metre.

SOLUTION

$\tan 60^\circ = \dfrac{h}{40}$, so $h = 40\sqrt{3} = 40 \times 1.732$

$h = 69.28$ m, which is 69 m to the nearest metre.

Example 3.2 A kite is flying at a height of 45 m. Its string makes an angle of 30° with the horizontal. Find the length of the string, assuming it is straight.

SOLUTION

$\sin 30^\circ = \dfrac{45}{l}$, so $l = \dfrac{45}{1/2}$

The string is 90 m long.

Example 3.3 From the top of a lighthouse 80 m high, the angle of depression of a boat is 45°. Find the distance of the boat from the foot of the lighthouse.

SOLUTION

The angle of elevation of the top from the boat is also 45°.

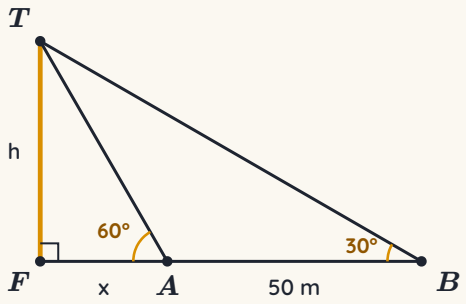
$\tan 45^\circ = \dfrac{80}{d}$

The boat is 80 m from the lighthouse.

4 Problems with two triangles

- Mark the common side, call the unknowns h and x, write one equation from each triangle and solve the two together.

Example 4.1 From a point on the ground, the angle of elevation of the top of a tower is 30°. On walking 50 m towards the tower, the angle of elevation becomes 60°. Find the height of the tower.



The tower FT seen from B (30°) and from A (60°)

SOLUTION

Let the height be h and the final distance from the tower be x.

$\tan 60^\circ = \dfrac{h}{x}$ gives $h = x\sqrt{3}$, and $\tan 30^\circ = \dfrac{h}{x+50}$ gives $x + 50 = h\sqrt{3}$

Substituting: $x + 50 = 3x$, so $x = 25$ and $h = 25\sqrt{3}$

Height $= 25\sqrt{3} \approx 43.3$ m

Example 4.2 Two ships are on opposite sides of a lighthouse 100 m high, in line with its foot. The angles of depression of the ships from the top are 30° and 45°. Find the distance between the ships.

SOLUTION

Ship 1: $\tan 30^\circ = \dfrac{100}{d_1}$, so $d_1 = 100\sqrt{3} = 173.2$ m

Ship 2: $\tan 45^\circ = \dfrac{100}{d_2}$, so $d_2 = 100$ m

Distance between the ships $= 173.2 + 100 = 273.2$ m

Example 4.3 From the top of a cliff 60 m high, the angles of depression of the top and the bottom of a tower are 30° and 60°. Find the height of the tower.

SOLUTION

Let the horizontal distance be d. Bottom of the tower: $\tan 60^\circ = \dfrac{60}{d}$, so $d = \dfrac{60}{\sqrt{3}} = 20\sqrt{3}$ m

Top of the tower: the cliff top is y m above the top of the tower, where $\tan 30^\circ = \dfrac{y}{d}$, so $y = \dfrac{20\sqrt{3}}{\sqrt{3}} = 20$ m

Height of the tower $= 60 - 20 = 40$ m

Example 4.4 A vertical pole and a vertical tower stand on level ground. From the top of the pole, the angle of elevation of the top of the tower is 60° and the angle of depression of its foot is 30°. If the tower is 50 m high, find the height of the pole.

SOLUTION

Let the pole be h m high and the distance be d. Depression of the foot: $\tan 30^\circ = \dfrac{h}{d}$, so $d = h\sqrt{3}$.

Elevation of the top: $\tan 60^\circ = \dfrac{50 - h}{d}$, so $50 - h = d\sqrt{3} = 3h$

$h = 12.5$ m

5 Steps for every problem

- Draw a clear figure and label all points.
- Mark the given angles from the horizontal.
- Write a trigonometric ratio for each right-angled triangle.
- Solve, then round off the final answer as asked.

6 Common mistakes

- Marking the angle of depression at the object instead of at the observer.
- When two objects are on **opposite** sides, add the distances; on the **same** side, subtract.
- Rounding $\sqrt{3}$ too early.
- Forgetting the observer's height when it is given.

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